

PROJECT DESIGN PHASE

SOLUTION ARCHITECTURE

ServiceNow Employee Data Import & Analytics Solution

Project Name	ServiceNow Employee Data Import & Analytics
Platform	ServiceNow
Core Technologies / Features	Import Sets, Transform Maps, Coalesce, Reports, Dashboards
Primary Use Case	Bulk employee data import, update, duplicate prevention and reporting

Purpose: Provide a clear architecture showing how employee data moves from an external spreadsheet into ServiceNow, how records are transformed and de-duplicated, and how the resulting data is exposed through reports and a centralized dashboard.

1. Solution Overview

The project implements an employee data management solution in ServiceNow using Import Sets and Transform Maps. External spreadsheet data is first loaded into an Import Set staging table. A Transform Map maps source fields to the Employee Test target table, after which the transformation creates or updates records. Coalesce is used on Employee ID to prevent duplicate records during repeated imports. Reports and an Employee Analytics Dashboard provide visibility into the resulting data.

Business / Technical Problem

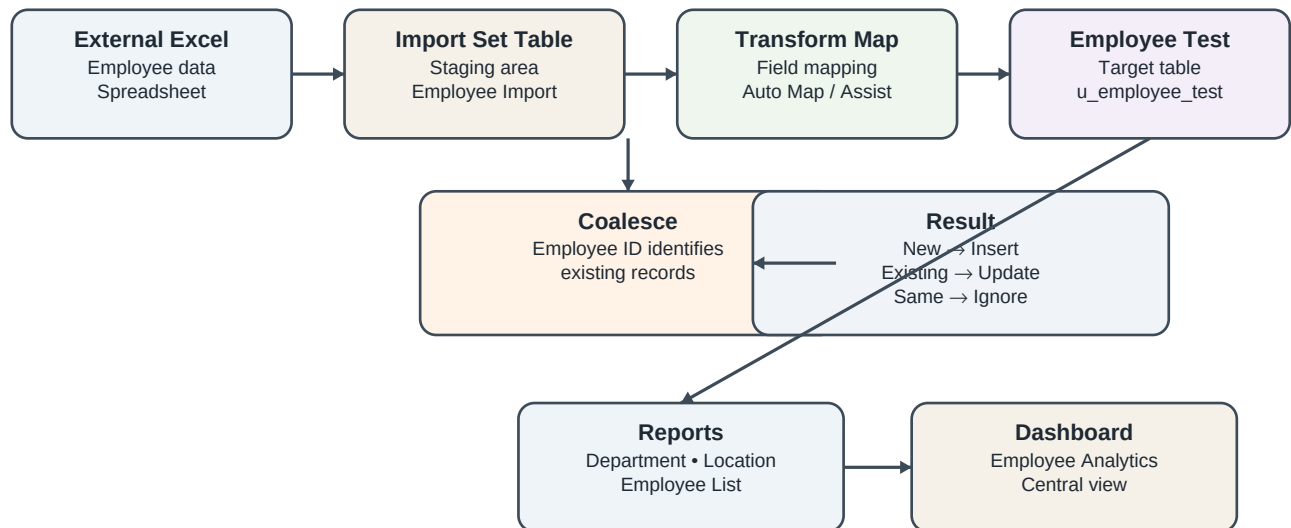
- Employee data is received in spreadsheet format and needs to be loaded into ServiceNow efficiently.
- Source fields must be mapped correctly to target employee fields.
- Repeated imports can create duplicate records unless an existing-record identifier is used.
- Administrators and HR users need a centralized view of employee data and trends.

Solution Objectives

- Create a controlled staging layer using an Import Set table.
- Map and transform spreadsheet fields into the Employee Test table.
- Use Coalesce with Employee ID to distinguish existing and new employee records.
- Validate inserted, updated and ignored records through Transform History.
- Provide Department, Location and Employee List reports and combine them in a dashboard.

2. Solution Architecture

Solution Architecture & Data Flow



Flow based on the uploaded ServiceNow Import Sets / Transform Maps project.

The architecture follows the project's documented flow: external Excel → Import Set staging → Transform Map → Employee Test target table → Reports → Dashboard. Coalesce operates during transformation to identify existing records and avoid duplicates.

3. Architecture Components

Component	Role in the solution
External Excel Spreadsheet	Source of structured employee data received for bulk import.
Employee Import (Import Set Table)	Temporary/staging area where source data is loaded before transformation.
Sample Spreadsheet Import (Transform Map)	Defines the source-to-target mapping and performs the transformation.
Employee Test (u_employee_test)	Custom target table containing Employee ID, Employee Name, Email, Department and Location.
Coalesce / Employee ID	Identifies existing employee records so repeated imports update rather than create duplicates.
Transform History	Shows import outcomes such as inserted, updated and ignored records.
Reports	Department-wise, location-wise and employee list views.
Employee Analytics Dashboard	Centralized visual view of the employee reports.

4. End-to-End Data Flow

1. Prepare source data — Create the employee spreadsheet and save it as an Excel file.

2. Load data — Load the spreadsheet into the Employee Import Import Set table.

- 3. Transform** — Use the Sample Spreadsheet Import Transform Map to map source fields to Employee Test.
- 4. Identify records** — Coalesce on Employee ID identifies whether an employee already exists.
- 5. Apply result** — New IDs are inserted; existing IDs with changed values are updated; repeated unchanged rows can be ignored.
- 6. Validate** — Review Transform History and the Employee Test table.
- 7. Report** — Create Employees by Department, Employees by Location and Employee List reports.
- 8. Dashboard** — Add the reports to Employee Analytics Dashboard for centralized monitoring.

5. Data Mapping

Source / Employee data	Target: Employee Test
Employee ID	Employee ID
Employee Name	Employee Name
Email	Email
Department	Department
Location	Location

6. Duplicate Prevention & Update Logic

The project enables Coalesce on a field map so that Employee ID is used to identify an existing record. The documented test demonstrates that when four rows are imported, two new employee IDs are inserted and two existing employee IDs are updated when their name/email values change. Re-uploading the same unchanged sheet results in ignored rows rather than duplicate records.

Condition	Documented transformation outcome
Employee ID does not exist	Insert new employee record
Employee ID exists and values changed	Update existing employee record
Same data imported again	Ignore unchanged rows

7. Reporting & Dashboard Architecture

- Employees by Department — Pie chart, grouped by Department, aggregation Count.
- Employees by Location — Bar chart, grouped by Location, aggregation Count.
- Employee List Report — List containing Employee ID, Employee Name, Email, Department and Location.
- Employee Analytics Dashboard — consolidates the three reports into one centralized view.

8. Expected / Documented Solution Outcome

- Employee records are imported from Excel into ServiceNow through a controlled Import Set and Transform Map flow.
- Source fields are mapped to the custom Employee Test table.
- Coalesce reduces duplicate creation during repeated imports and supports updates to existing records.
- Transform History provides visibility into the import results.
- Reports and a dashboard provide centralized visibility into employee data.

9. Architecture Summary

This solution bridges the business requirement of efficient employee-data migration with ServiceNow platform capabilities. The architecture separates staging, transformation, target storage and reporting, while Coalesce adds a record-matching mechanism for repeated imports. The resulting reports and dashboard provide a single place for administrators and HR teams to view employee information.

Source basis: uploaded project document “Import Data using Transform Maps (Spreadsheet)” and the supplied “Solution Architecture” project-design template.